

Task 11: Phishing Attack Simulation & Detection

Tool Used:

- **GoPhish** – phishing simulation framework
- **Gmail SMTP** – to send phishing emails
- **VirtualBox** – to run Windows as victim machine

1. Introduction

Phishing is a type of social engineering attack where attackers send fake emails that look real to trick users into clicking links or giving personal information.

In this experiment, I used **GoPhish** to simulate a phishing attack in a controlled lab environment.

The aim was to understand:

- How phishing emails are created
- How users get tricked
- How attackers track victims
- And how such attacks can be detected and prevented

2. Lab Setup

System	Purpose
Ubuntu (Host OS)	Attacker system – GoPhish server
Windows (VirtualBox VM)	Victim system

The phishing server was hosted on Ubuntu, and the victim email was opened inside the Windows virtual machine. This setup simulates a real-world phishing scenario.

4. Step-by-Step Process

Step 1: Understanding Phishing

Phishing attacks use urgency, fake identity, and suspicious links to trick users into clicking.

Step 2: Creating Fake Email Template

A phishing email was created with a fake IT warning message and a tracking link using:

{{ .URL }}

This automatically inserts the phishing link for each user.

Screenshot : Email Template Created :

The screenshot shows the Gophish web interface for creating an email template. On the left is a sidebar with navigation links: Dashboard, Campaigns, Users & Groups, Email Templates (selected), Landing Pages, Sending Profiles, Account Settings, User Management (Admin), Webhooks (Admin), User Guide, and API Documentation. The main area is titled 'Email Template' and contains a '+ New Template' button and a message 'No templates yet. Let's create one'. A modal window is open for creating a new template. It has fields for 'Envelope Sender' (IT Support <it@company.com>), 'Subject' (Account Verification Required), and tabs for 'Text' and 'HTML' (selected). The HTML content area contains a phishing message: 'Dear User, Your email account is scheduled for suspension due to suspicious activity. Please verify your account immediately: {{.URL}} IT Support Team'. Below the content area are checkboxes for 'Add Tracking Image' (checked) and a '+ Add Files' button. At the bottom of the modal are 'Cancel' and 'Save Template' buttons. A table at the bottom of the modal shows 'No data available in table' and 'Showing 0 to 0 of 0 entries'.

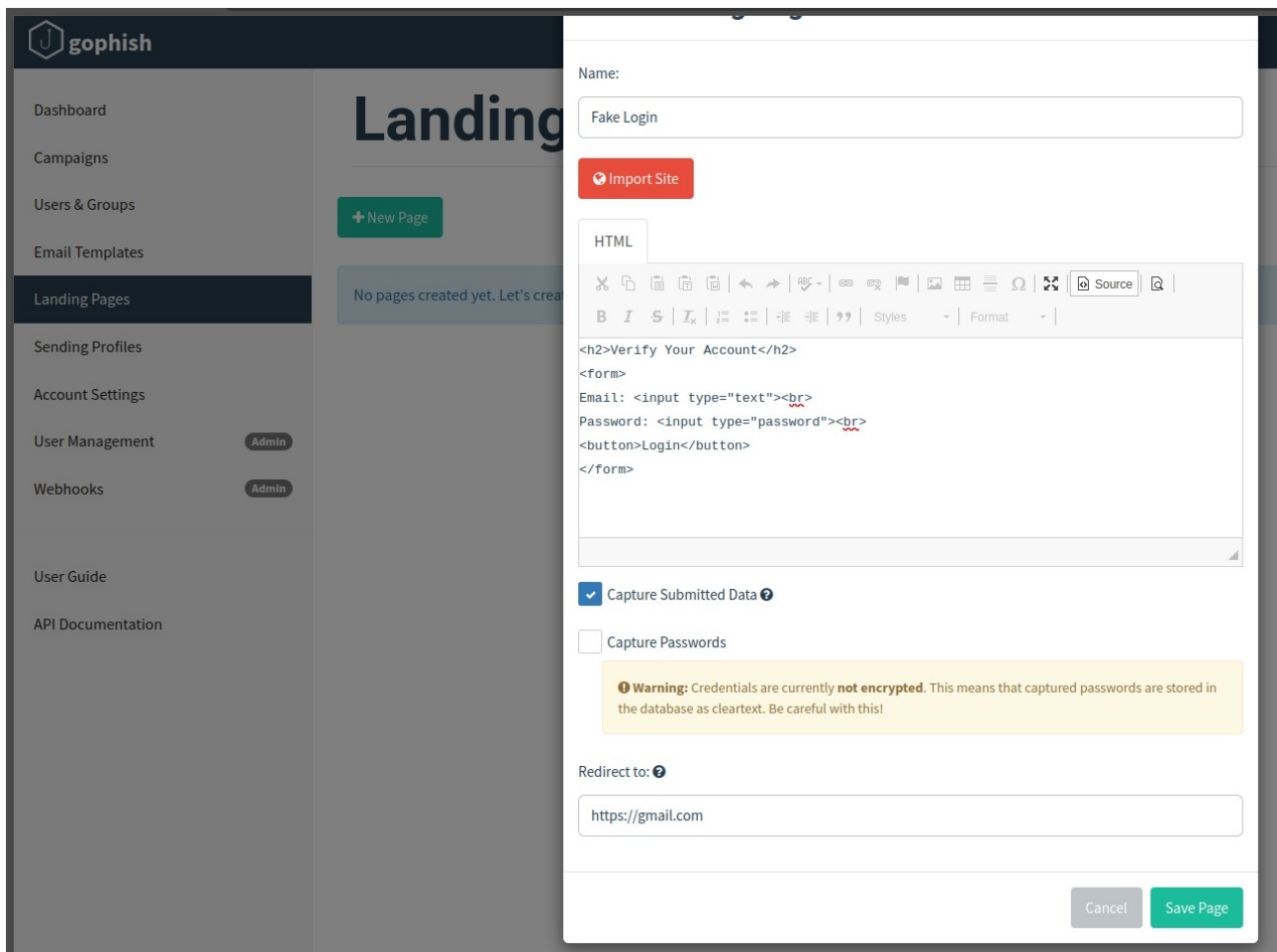
Step 3: Creating Fake Landing Page

A fake login page was created that asked for:

- Email
- Password

Only “**Capture Submitted Data**” was enabled.
Password capture was disabled for ethical reasons.

Screenshot : Fake Landing Page:



Step 4: Sending Profile Setup

A Gmail account with an **App Password** was used to send emails through SMTP.

Screenshot : Sending Profile:

gophish

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

+ New Profile

No profiles created yet. Let's create one.

Name:

Gmail Sender

Interface Type:

SMTP

SMTP From: ?

IT Support <mounika4p@gmail.com>

Host:

smtp.gmail.com

Username:

mounika4p@gmail.com

Password:

☒ Ignore Certificate Errors ?

Email Headers:

X-Custom-Header

{{.URL}}-gophish

+ Add Custom Header

Show 10 entries

Search:

Header

Value

No data available in table

Showing 0 to 0 of 0 entries

Send Test Email

Cancel

Save Profile

Step 5: Adding Victim User

A test email ID was added as the victim and opened inside the Windows VM.

Screenshot : Victim User Group:

gophish

admin

Dashboard

Campaigns

Users & Groups

Email Templates

Landing Pages

Sending Profiles

Account Settings

User Management

Webhooks

User Guide

API Documentation

+ New Group

Group added successfully!

Show 10 entries

Search:

Name	# of Members	Modified Date
Victims	1	February 2nd 2026, 5:55:50 pm

Showing 1 to 1 of 1 entries

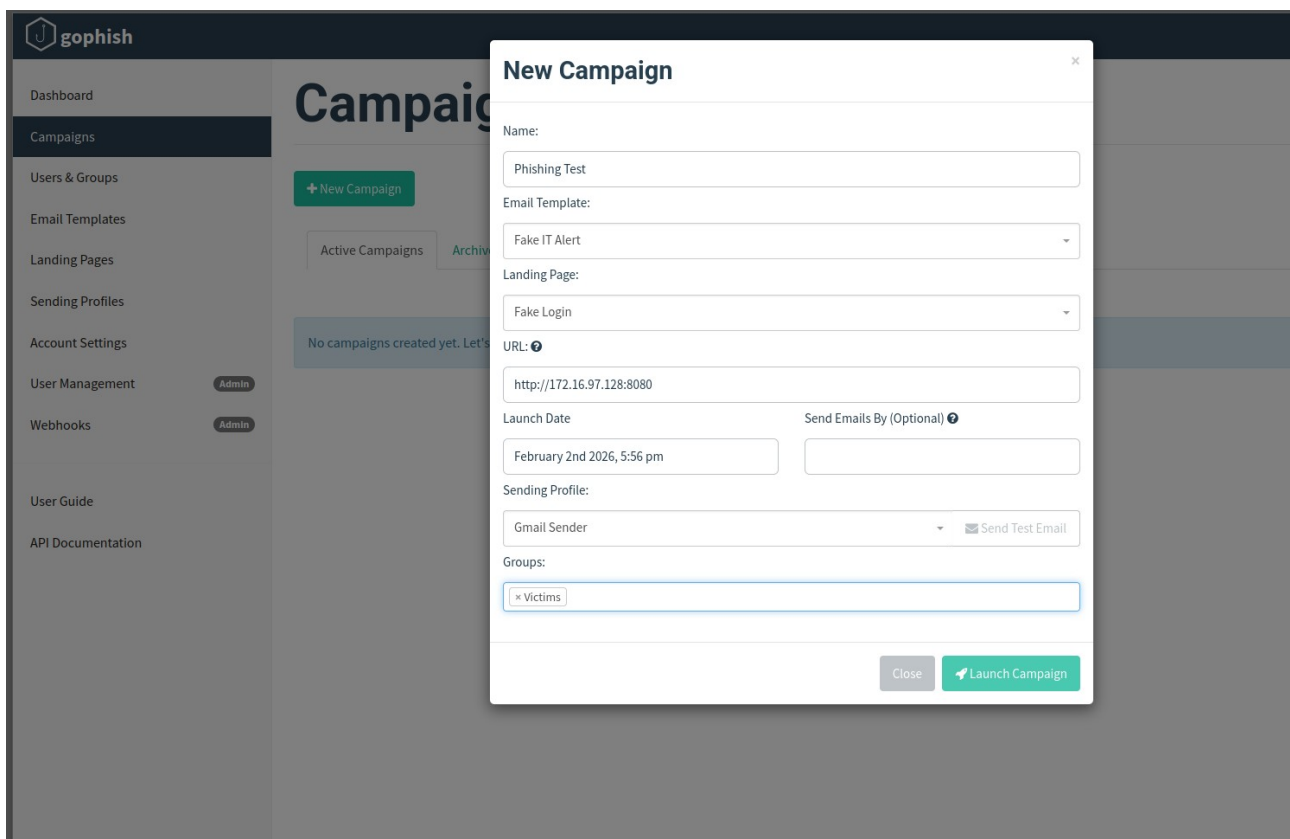
Previous1Next

Step 6: Launching Campaign

The campaign was launched with:

- Email template
- Landing page
- Victim user
- Ubuntu IP link

Screenshot : Campaign setup:



The screenshot shows the 'gophish' web interface with a 'New Campaign' modal form open. The form contains the following fields and options:

- Name:** Text input field containing 'Phishing Test'.
- Email Template:** Dropdown menu with 'Fake IT Alert' selected.
- Landing Page:** Dropdown menu with 'Fake Login' selected.
- URL:** Text input field containing 'http://172.16.97.128:8080'.
- Launch Date:** Text input field containing 'February 2nd 2026, 5:56 pm'.
- Send Emails By (Optional):** Empty text input field.
- Sending Profile:** Dropdown menu with 'Gmail Sender' selected, and a 'Send Test Email' button.
- Groups:** Text input field containing 'x Victims'.

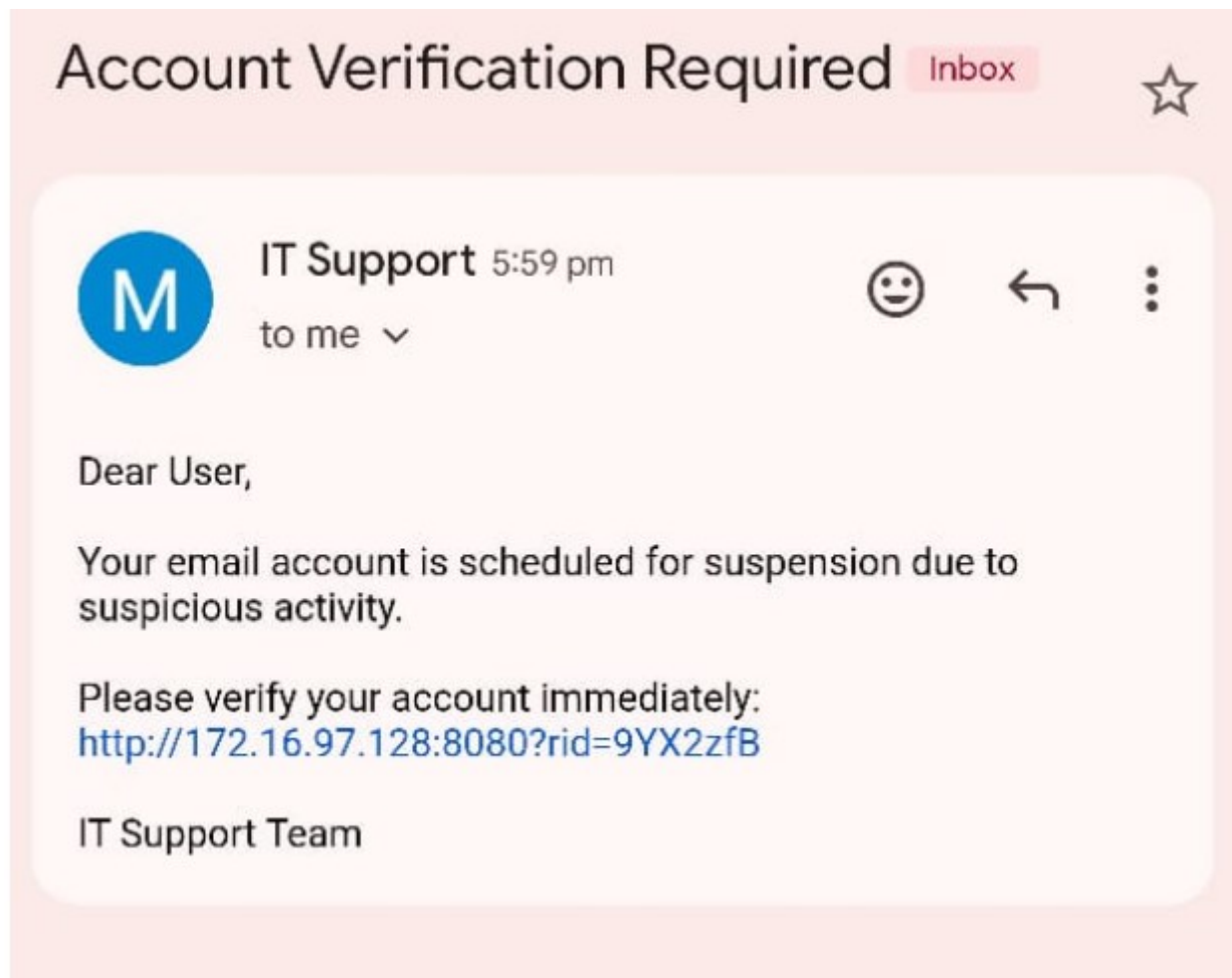
At the bottom of the modal are two buttons: 'Close' and 'Launch Campaign'.

Step 7: Victim Interaction

The victim:

1. Received the phishing email
2. Clicked the malicious link
3. Entered fake credentials
4. Got redirected (error due to NAT/network, which is expected in a lab)

Screenshot : Phishing email in inbox :



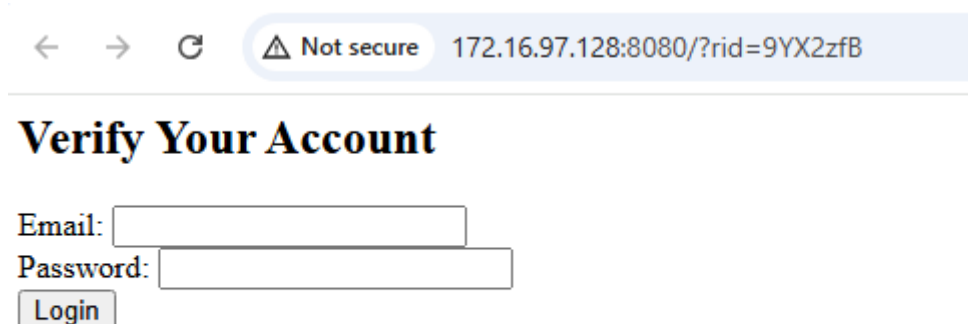
Step 8: Tracking Results

GoPhish successfully recorded:

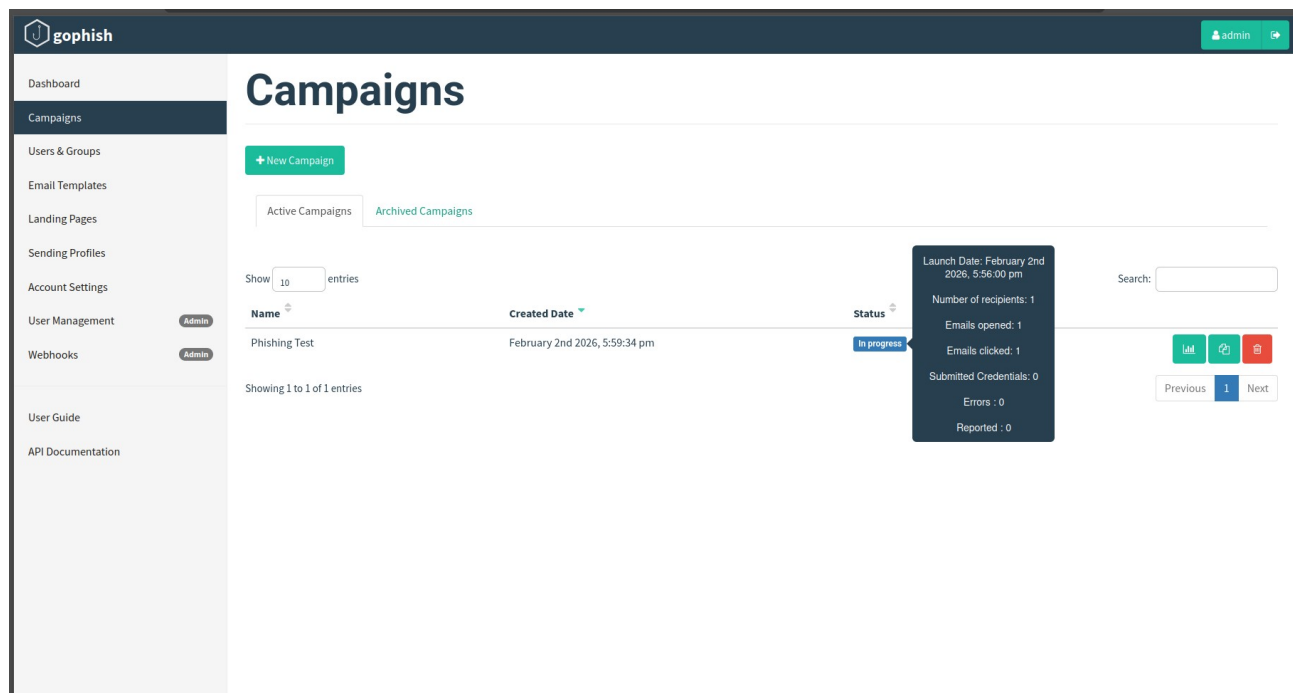
- Email sent
- Link clicked
- Data submitted

This confirmed the phishing attack worked.

Screenshot : Fake login page:



Screenshot: GoPhish Campaign Results Dashboard :



5. Red Flags Observed

- Urgent message (“account suspension”)
- Fake IT identity
- Suspicious IP link
- No personalization
- No official domain

6. Prevention Methods

- User awareness training
- Do not click unknown links
- Verify sender address
- Enable Multi-Factor Authentication (MFA)
- Use email filtering systems

7. Conclusion

This experiment showed how easy it is to trick users using social engineering. The simulation helped me understand how phishing works, how attackers track victims, and why user awareness is important for cybersecurity.